Lab No. 6

Set Up Nexus Repository and Configure User Access

NAME: LAIBA RANI

REGISTRATION NO.: 2023-BCS-053

COURSE CODE: CS-649

SECTION: A

GITHUB USERNAME: laibarani899-cpu

Task 1: Run Nexus Repository and Complete the Initial Setup

```
laibarani899-cpu@ubuntu:~$ docker logs -f nexus
2026-11-09 16:46:04,525+0000 INFO  [main] *SYSTEM com.sonatype.nexus.bootstrap.entrypoint.pro.SonatypeNexusRepositoryApplication - St
arting SonatypeNexusRepositoryApplication v3.98.1-01 using Java 21.0.13.0.3 with PID 1 (/opt/sonatype/nexus/bin/sonatype-nexus-reposi
tory-3.98.1-01.jar started by nexus in /opt/sonatype/nexus)
2026-11-09 16:46:04,532+0000 INFO  [main] *SYSTEM com.sonatype.nexus.bootstrap.entrypoint.pro.SonatypeNexusRepositoryApplication - No
 active profile set, falling back to 1 default profile: "default"
2026-11-09 16:46:10,365+0000 INFO  [main] *SYSTEM java.util.prefs - Created user preferences directory.
2026-11-09 16:46:10,535+0000 INFO  [main] *SYSTEM org.sonatype.nexus.bootstrap.entrypoint.ApplicationLauncher - Starting nexus with e
dition COMMUNITY
2026-11-09 16:46:10,847+0000 INFO  [main] *SYSTEM org.sonatype.nexus.bootstrap.entrypoint.SpringComponentScan - Scanning for componen
ts in packages: [org.sonatype.nexus, com.sonatype.nexus]
2026-11-09 16:47:55,786+0000 INFO  [main] *SYSTEM org.sonatype.nexus.quartz.internal.QuartzSchedulerProvider - Thread-pool size: 20,
Thread-pool priority: 5
2026-11-09 16:47:57,020+0000 INFO  [main] *SYSTEM org.sonatype.nexus.internal.log.overrides.file.LogbackLoggerOverrides - File: /nexu
s-data/etc/logback/logback-overrides.xml
2026-11-09 16:47:57,146+0000 INFO  [main] *SYSTEM org.apache.shiro.nexus.NexusWebSessionManager - Global session timeout: 1800000 ms
2026-11-09 16:47:57,149+0000 INFO  [main] *SYSTEM org.apache.shiro.nexus.NexusWebSessionManager - Session-cookie prototype: name=NXSE
SSIONID, secure=true
2026-11-09 16:47:57,778+0000 INFO  [main] *SYSTEM org.sonatype.nexus.repository.aws.ecr.auth.store.EcrTokenStoreDAO - EcrTokenStoreDA
O bean initialised (shared module)
2026-11-09 16:47:58,836+0000 WARN  [main] *SYSTEM org.sonatype.nexus.repository.oci.internal.security.OciCursorSigner - OCI cursor si
gning: SecretStore is unavailable (nexus.jwt.enabled=false). Cursor validation will fail across cluster nodes. Enable JWT for HA depl
oyments.
2026-11-09 16:47:58,922+0000 INFO  [main] *SYSTEM org.sonatype.nexus.repository.aws.ecr.auth.EcrTokenManager - EcrTokenManager bean i
nitialised (shared module)
2026-11-09 16:47:58,927+0000 INFO  [main] *SYSTEM org.sonatype.nexus.repository.aws.ecr.auth.EcrAuthenticationHandler - EcrAuthentica
tionHandler bean initialised (shared module)
2026-11-09 16:47:59,144+0000 INFO  [main] *SYSTEM org.sonatype.nexus.cache.internal.ehcache.EhCacheManagerProvider - Creating cache-m
anager with configuration: file:/opt/sonatype/nexus/etc/fabric/ehcache.xml
2026-11-09 16:48:07,815+0000 INFO  [main] *SYSTEM org.ehcache.jsr107.ConfigurationMerger - Configuration of cache ansiblegalaxy-impor
```

Verify that Nexus is running

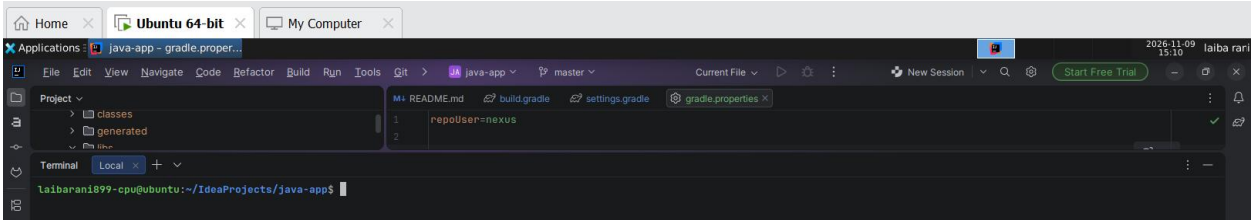
Task 2: Create a Role and User and Verify Access

```
laibarani899-cpu@ubuntu:~$ curl -i -u nexus http://192.168.88.130:8006/service/rest/v1/status
Enter host password for user 'nexus':
HTTP/1.1 200 OK
Server: Nexus/3.98.1-01 (COMMUNITY)
Vary: Accept-Encoding
X-Content-Type-Options: nosniff
Content-Security-Policy: default-src http: data: blob: 'unsafe-inline'; script-src http: 'unsafe-inline' 'unsafe-eval'
Content-Length: 0

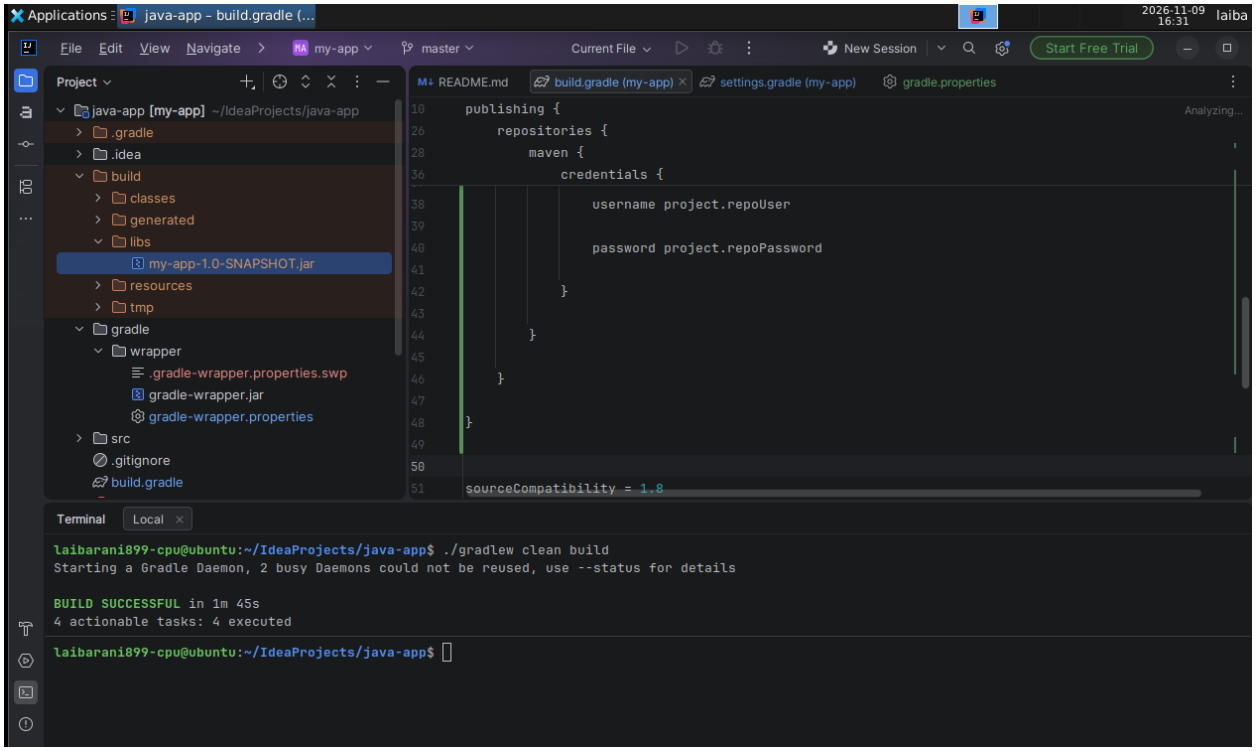
laibarani899-cpu@ubuntu:~$
```

Verify the user's access

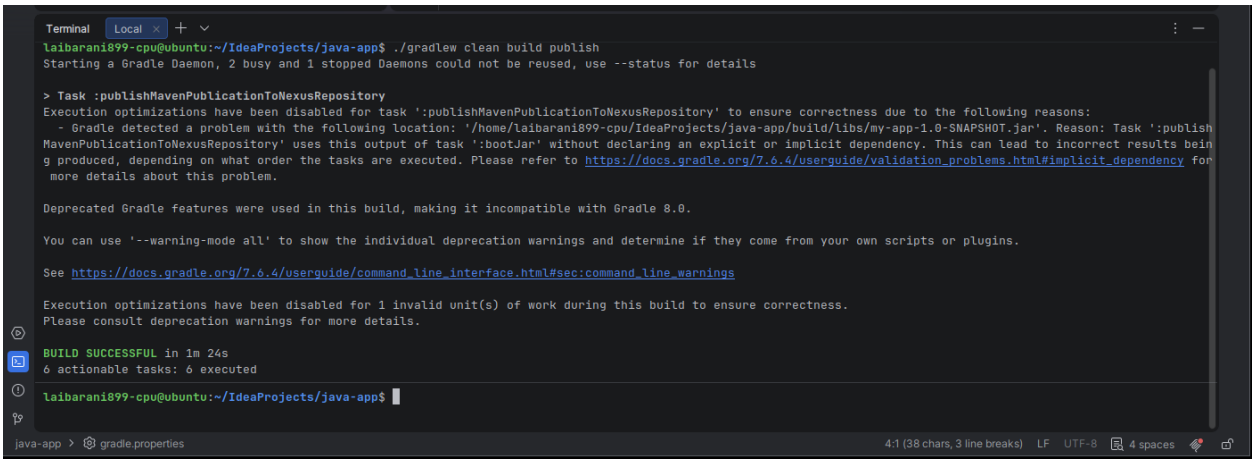
Task 3: Build and Publish the Gradle Application to Nexus



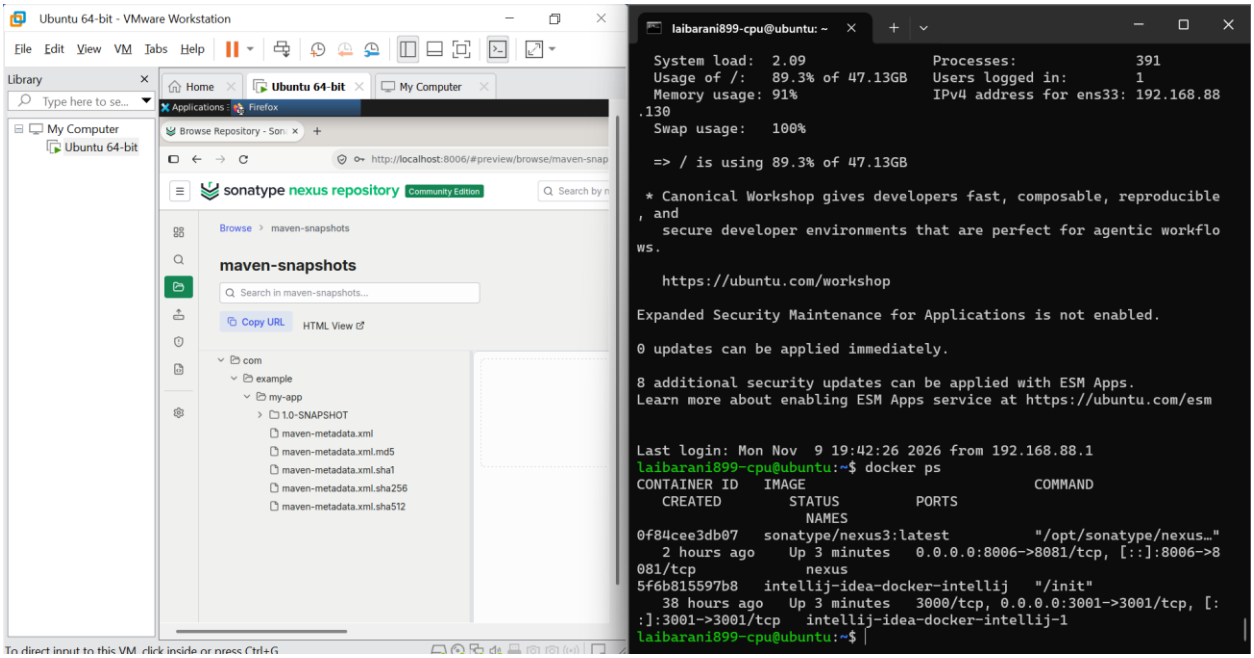
Configure `build.gradle` for Maven publication



Build the JAR

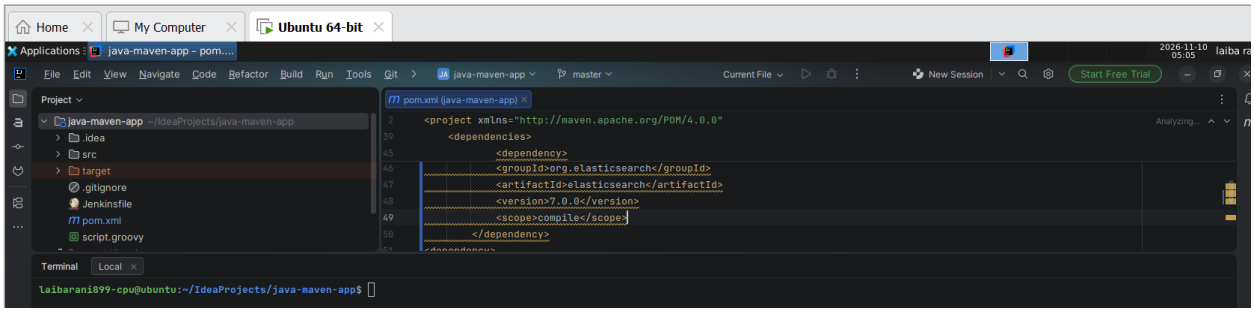


Publish the JAR to Nexus

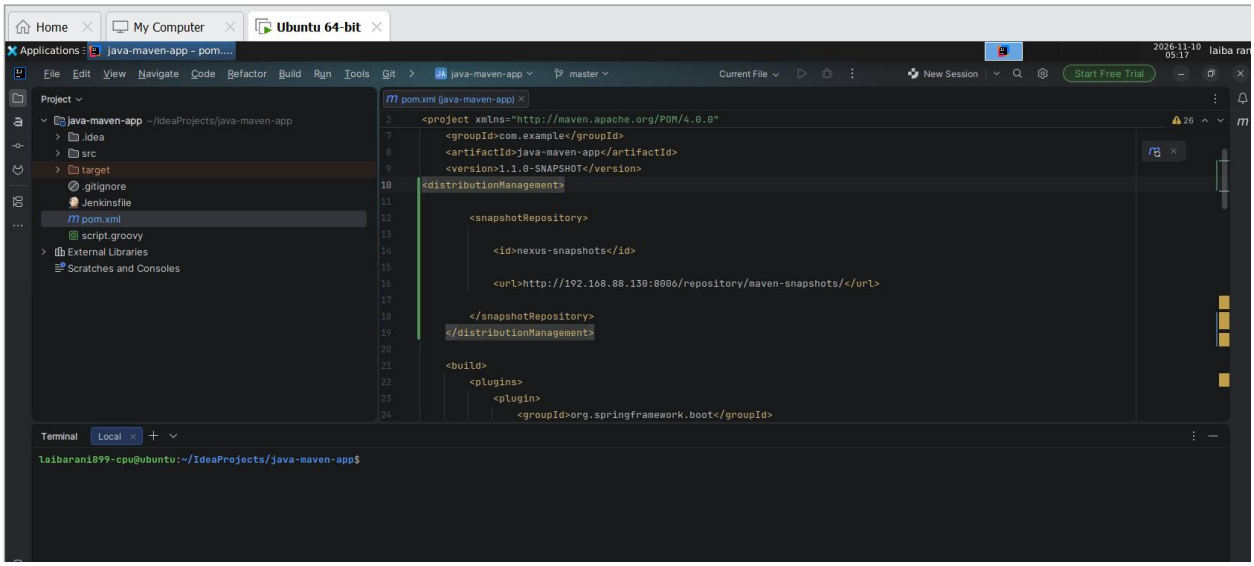


Verify the artifact in Nexus

Task 4: Publish the Maven Application to Nexus



Clone or reopen the Maven project



Add snapshot distribution management to `pom.xml`



Configure the Nexus deployment credentials

```
Terminal  Local

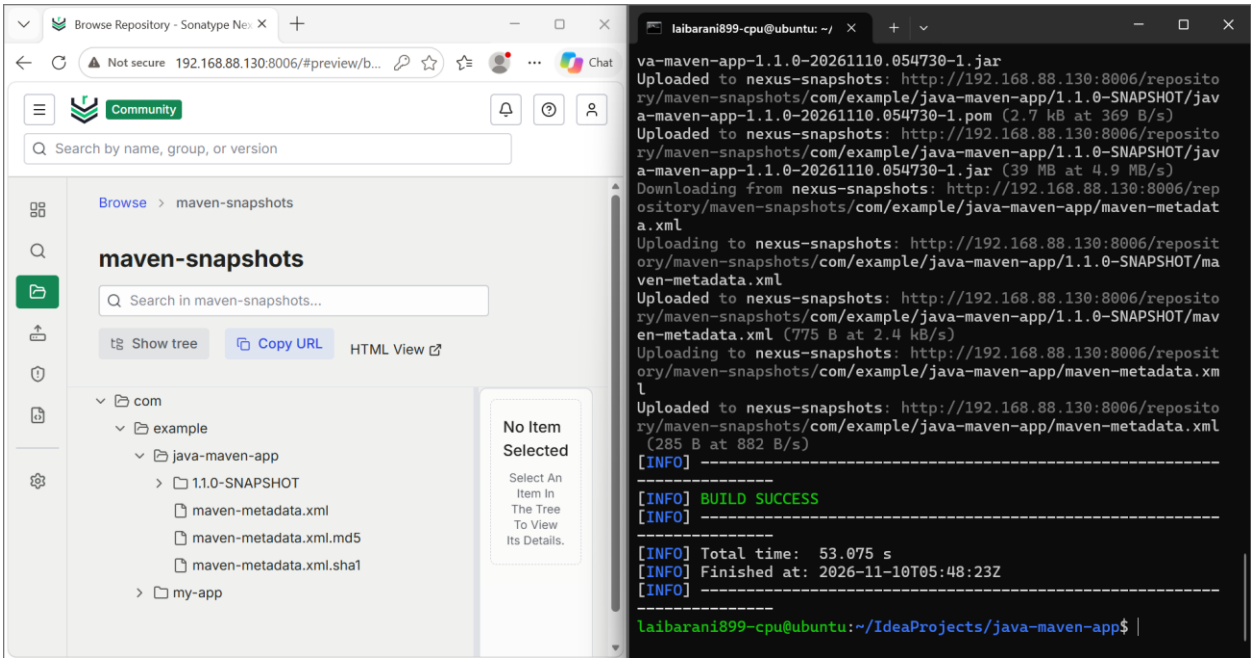
laibarani899-cpu@ubuntu:~/IdeaProjects/java-maven-app$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:java-maven-app >--------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/laibarani899-cpu/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO] Building jar: /home/laibarani899-cpu/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.430 s
[INFO] Finished at: 2026-11-10T05:36:56Z
[INFO] ------------------------------------------------------------------------

laibarani899-cpu@ubuntu:~/IdeaProjects/java-maven-app$
```

Package the Maven application

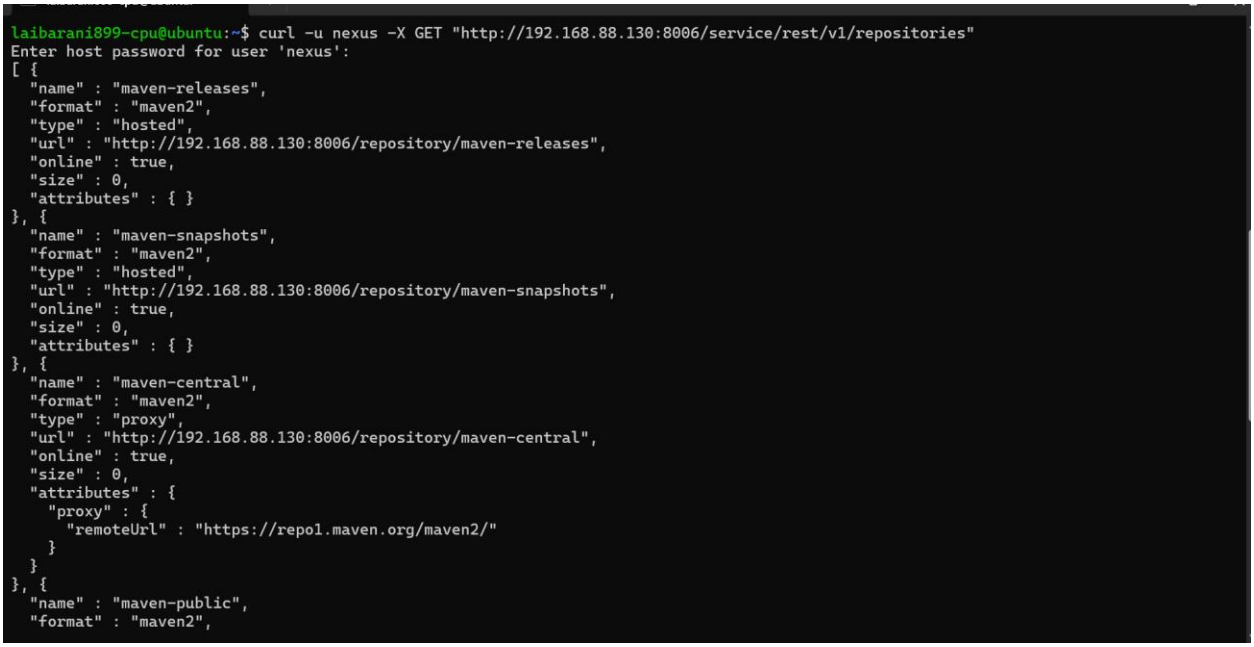
```
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 505 kB/s)
[INFO] Installing /home/laibarani899-cpu/IdeaProjects/java-maven-app/pom.xml to /home/laibarani899-cpu/.m2/repository/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-SNAPSHOT.pom
[INFO] Installing /home/laibarani899-cpu/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar to /home/laibarani899-cpu/.m2/repository/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- deploy:3.1.2:deploy (default-deploy) @ java-maven-app ---
Downloading from nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/maven-metadata.xml
Uploading to nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261110.054730-1.pom
Uploading to nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261110.054730-1.jar
Uploaded to nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261110.054730-1.pom (2.7 kB at 369 B/s)
Uploaded to nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/java-maven-app-1.1.0-20261110.054730-1.jar (39 MB at 4.9 MB/s)
Downloading from nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/maven-metadata.xml
Uploading to nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/maven-metadata.xml
Uploaded to nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/1.1.0-SNAPSHOT/maven-metadata.xml (775 B at 2.4 kB/s)
Uploading to nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/maven-metadata.xml
Uploaded to nexus-snapshots: http://192.168.88.130:8006/repository/maven-snapshots/com/example/java-maven-app/maven-metadata.xml (285 B at 882 B/s)
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  53.075 s
[INFO] Finished at: 2026-11-10T05:48:23Z
[INFO] ------------------------------------------------------------------------
laibarani899-cpu@ubuntu:~/IdeaProjects/java-maven-app$
```

Deploy the snapshot to Nexus



Verify the Maven artifact in Nexus

Task 5: Inspect Nexus Repositories and Components Using the REST API



List repositories as the nexus user

```
} ]laibarani899-cpu@ubuntu:curl -u admin -X GET "http://192.168.88.130:8006/service/rest/v1/repositories"
Enter host password for user 'admin':
[ {
  "name" : "maven-releases",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.88.130:8006/repository/maven-releases",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "maven-snapshots",
  "format" : "maven2",
  "type" : "hosted",
  "url" : "http://192.168.88.130:8006/repository/maven-snapshots",
  "online" : true,
  "size" : 0,
  "attributes" : { }
}, {
  "name" : "maven-central",
  "format" : "maven2",
  "type" : "proxy",
  "url" : "http://192.168.88.130:8006/repository/maven-central",
  "online" : true,
  "size" : 0,
  "attributes" : {
    "proxy" : {
      "remoteUrl" : "https://repo1.maven.org/maven2/"
    }
  }
}, {
  "name" : "maven-public",
  "format" : "maven2",
```

List repositories as the `admin` user

```
laibarani899-cpu@ubuntu:~$ curl -u nexus -X GET "http://192.168.88.130:8006/service/rest/v1/components?repository=maven-snapshots"
Enter host password for user 'nexus':
{
  "items" : [ {
    "id" : "bWF2ZW4tc25hcHNob3RzOjRmMWJiY2Rk",
    "repository" : "maven-snapshots",
    "format" : "maven2",
    "group" : "com.example",
    "name" : "my-app",
    "version" : "1.0-20261109.194724-1",
    "assets" : [ {
      "downloadUrl" : "http://192.168.88.130:8006/repository/maven-snapshots/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261109.194724-1.jar",
      "path" : "/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261109.194724-1.jar",
      "id" : "bWF2ZW4tc25hcHNob3RzOmRhYTY2ZDJl",
      "repository" : "maven-snapshots",
      "format" : "maven2",
      "checksum" : {
        "md5" : "baefb2cabb5bf3278b281527e0c59b5f",
        "sha512" : "b3d5fe1739d5313e1396a2072db7612752439e999ab45147853f0dcfb17f09ca8e86622eb6864f36f60a6cde2588afeed897ebbad56ebe936c018cfdabeedb9f",
        "sha1" : "eb56fd9638db99795cee03fd0e4e01989476a8c3",
        "sha256" : "31855f920d0b4cd7231b3211d8f6f78610af097daa7c29d63ca99e545b46d8bb"
      },
      "contentType" : "application/java-archive",
      "lastModified" : "2026-11-09T19:48:20.692+00:00",
      "lastDownloaded" : null,
      "uploader" : "nexus",
      "uploaderIp" : "192.168.88.130",
      "fileSize" : 39682480,
      "blobCreated" : "2026-11-09T19:48:20.770+00:00",
      "blobStoreName" : "default",
```

List the components in `maven-snapshots`

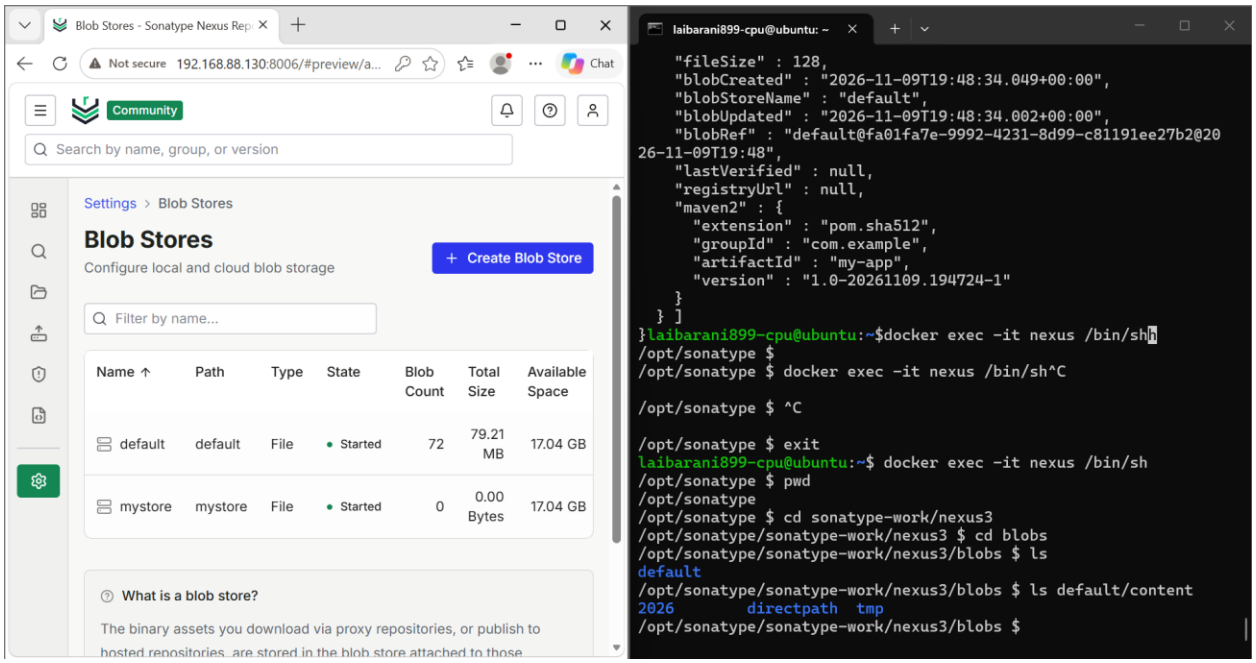
```
laibarani899-cpu@ubuntu:~$  curl -u nexus -X GET "http://192.168.88.130:8006/service/rest/v1/components/bWF2ZW4tc25hcHNob3RzOjRmMWJiY
2Rk"
Enter host password for user 'nexus':
{
  "id" : "bWF2ZW4tc25hcHNob3RzOjRmMWJiY2Rk",
  "repository" : "maven-snapshots",
  "format" : "maven2",
  "group" : "com.example",
  "name" : "my-app",
  "version" : "1.0-20261109.194724-1",
  "assets" : [ {
    "downloadUrl" : "http://192.168.88.130:8006/repository/maven-snapshots/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261109.194724
-1.jar",
    "path" : "/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261109.194724-1.jar",
    "id" : "bWF2ZW4tc25hcHNob3RzOmRhYTY2ZDJl",
    "repository" : "maven-snapshots",
    "format" : "maven2",
    "checksum" : {
      "md5" : "baefb2cabb5bf3278b281527e0c59b5f",
      "sha512" : "b3d5fe1739d5313e1396a2072db7612752439e999ab45147853f0dcfb17f09ca8e86622eb6864f36f60a6cde2588afeed897ebbad56ebe936c0
18cfdabeedb9f",
      "sha1" : "eb56fd9638db99795cee03fd0e4e01989476a8c3",
      "sha256" : "31855f920d0b4cd7231b3211d8f6f78610af097daa7c29d63ca99e545b46d8bb"
    },
    "contentType" : "application/java-archive",
    "lastModified" : "2026-11-09T19:48:20.692+00:00",
    "lastDownloaded" : null,
    "uploader" : "nexus",
    "uploaderIp" : "192.168.88.130",
    "fileSize" : 39682480,
    "blobCreated" : "2026-11-09T19:48:20.770+00:00",
```

Get one component's details by ID

Task 6: Create a Nexus Blob Store and Docker Hosted Repository

```
laibarani899-cpu@ubuntu:~$ docker exec -it nexus /bin/sh
/opt/sonatype $ pwd
/opt/sonatype
/opt/sonatype $ cd sonatype-work/nexus3
/opt/sonatype/sonatype-work/nexus3 $ cd blobs
/opt/sonatype/sonatype-work/nexus3/blobs $ ls
default
/opt/sonatype/sonatype-work/nexus3/blobs $ ls default/content
2026        directpath  tmp
/opt/sonatype/sonatype-work/nexus3/blobs $
```

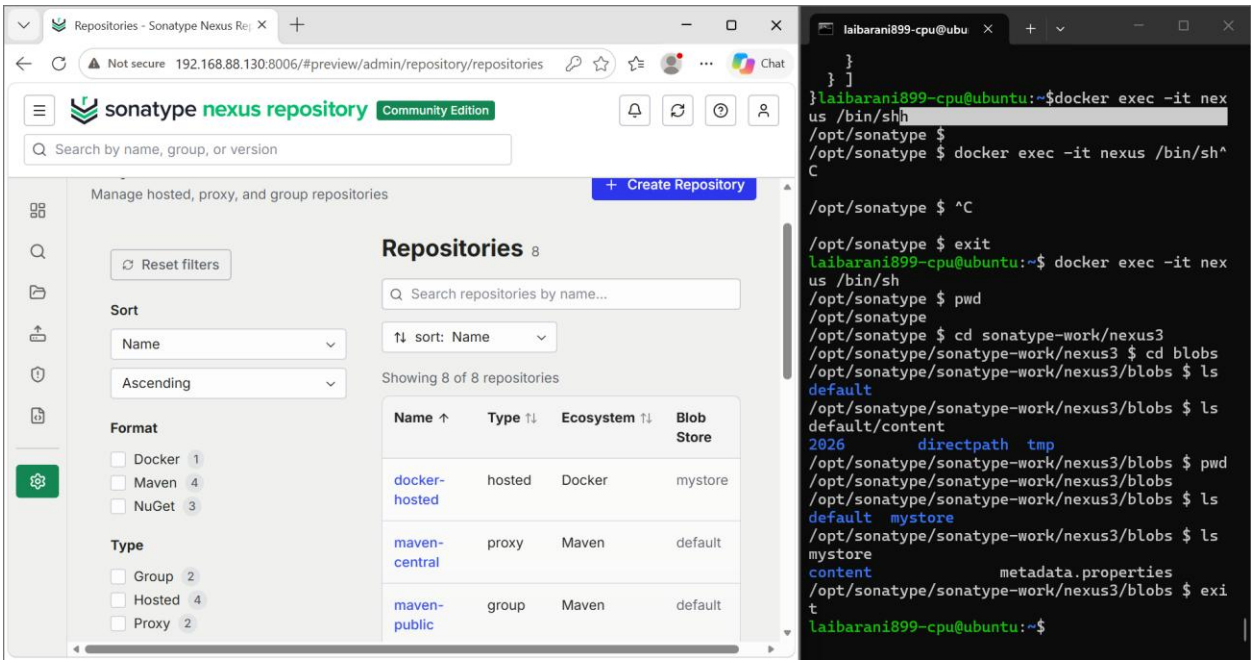
Inspect the default blob-store directory inside the Nexus container



Create the `mystore` file blob store

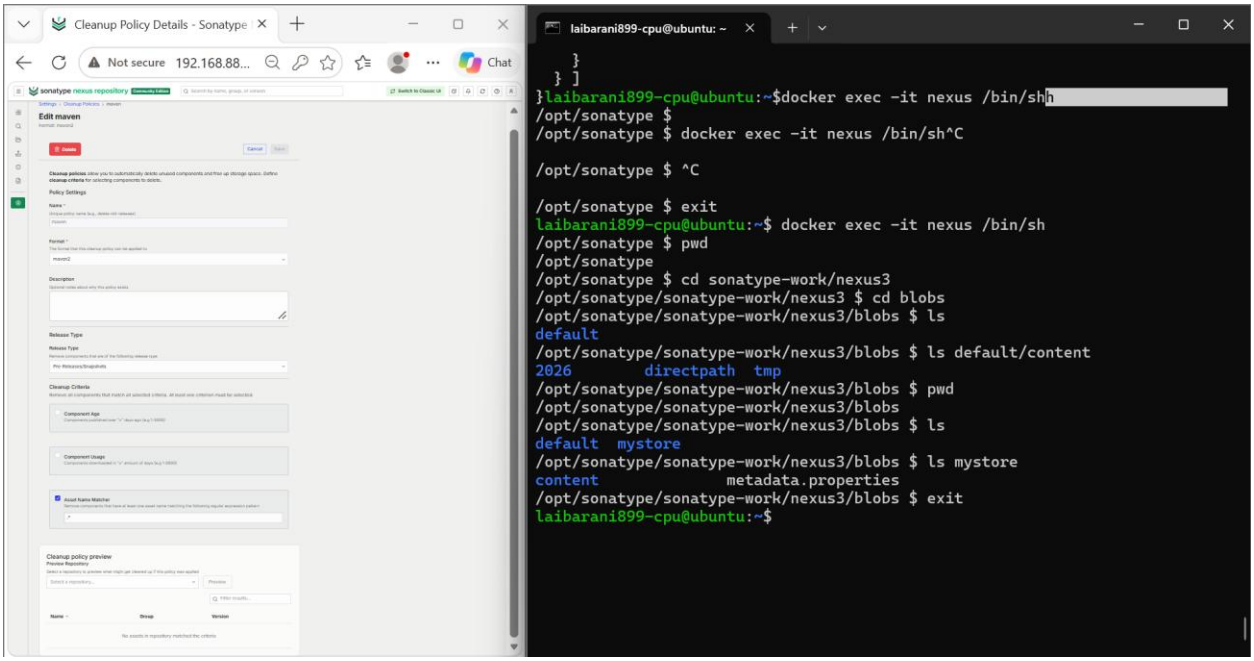
```
/opt/sonatype/sonatype-work/nexus3/blobs $ pwd
/opt/sonatype/sonatype-work/nexus3/blobs
/opt/sonatype/sonatype-work/nexus3/blobs $ ls
default  mystore
/opt/sonatype/sonatype-work/nexus3/blobs $ ls mystore
content                 metadata.properties
/opt/sonatype/sonatype-work/nexus3/blobs $ exit
laibarani899-cpu@ubuntu:~$
```

Verify the new blob-store directory inside the existing container shell

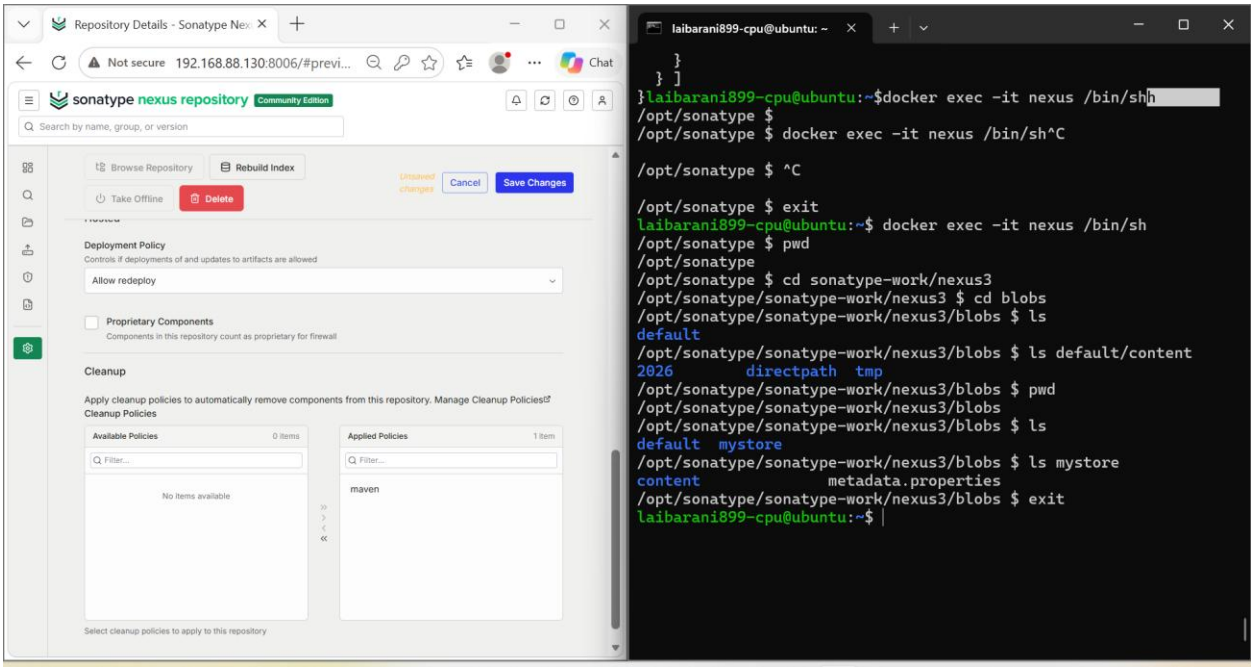


Create a Docker hosted repository that uses `mystore`

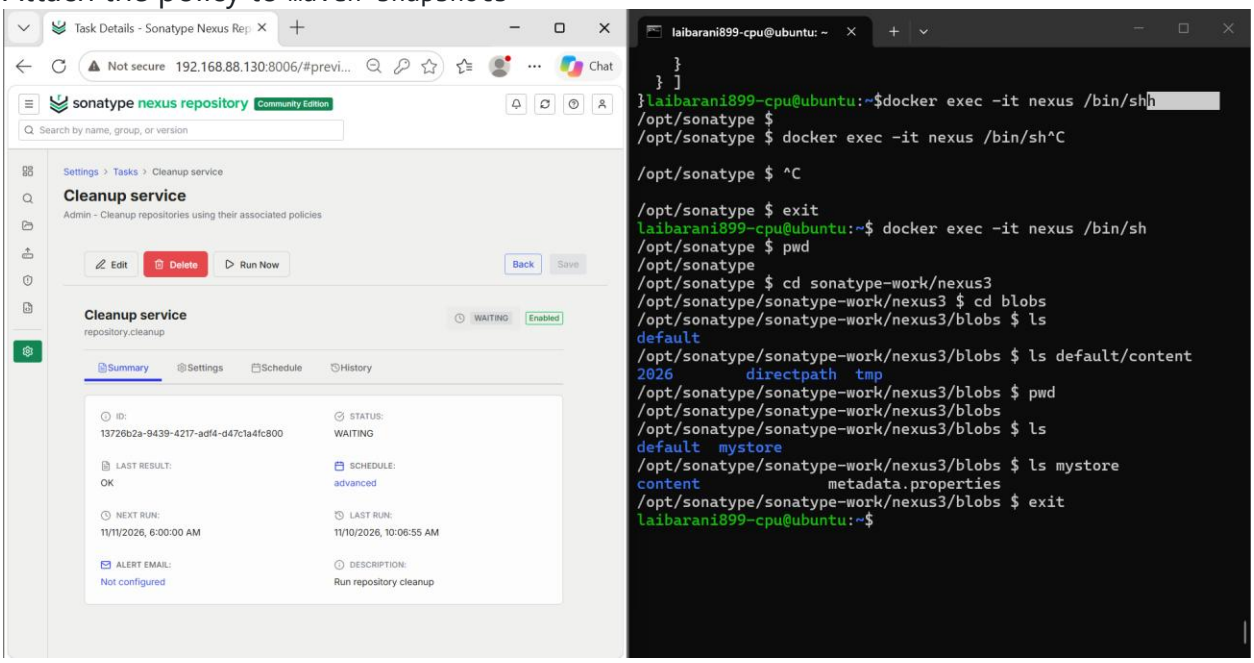
Task 7: Create and Run a Maven Snapshot Cleanup Policy



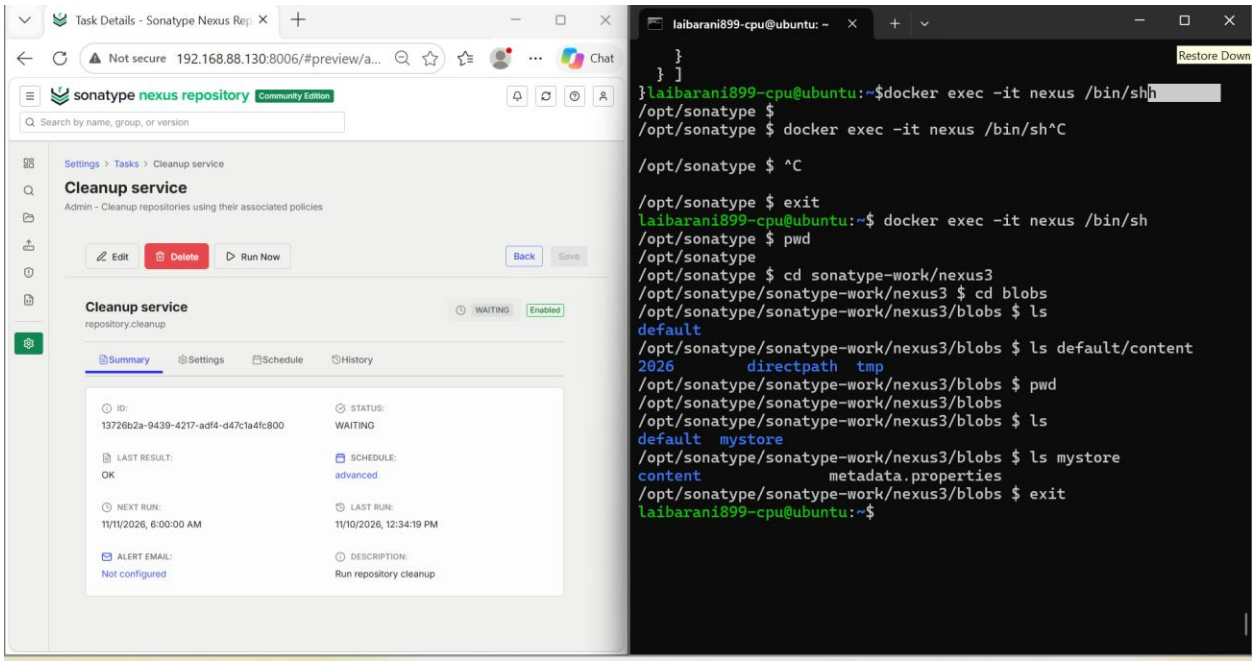
Create the Maven cleanup policy



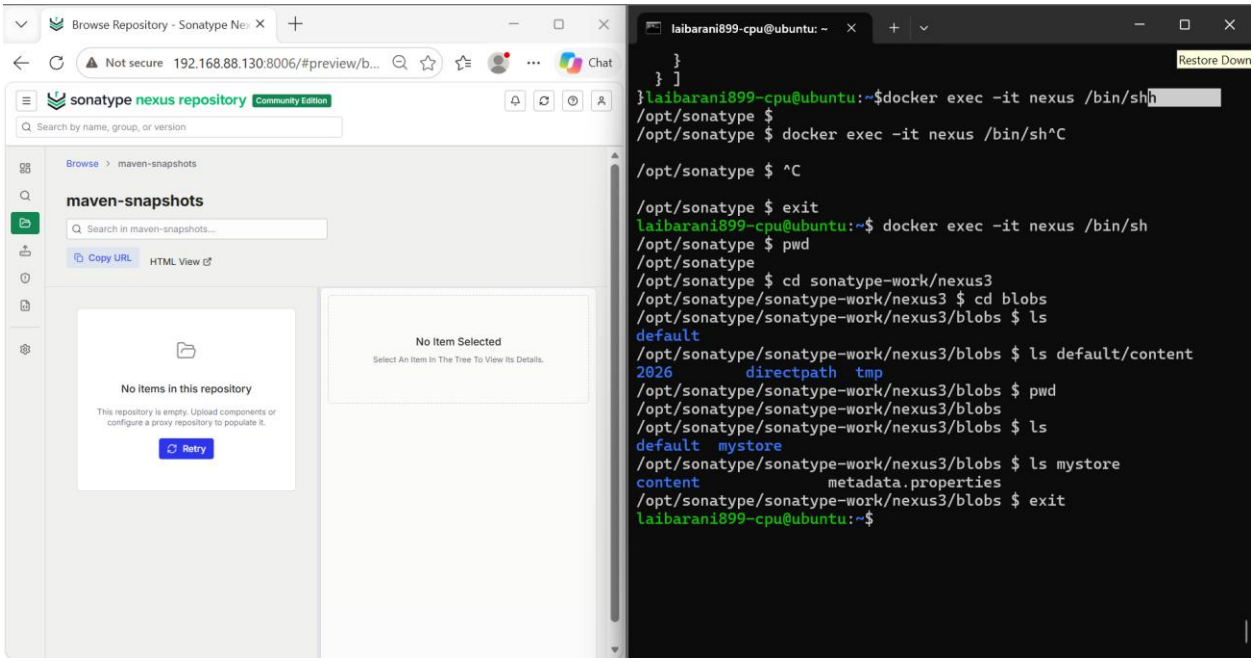
Attach the policy to `maven-snapshots`



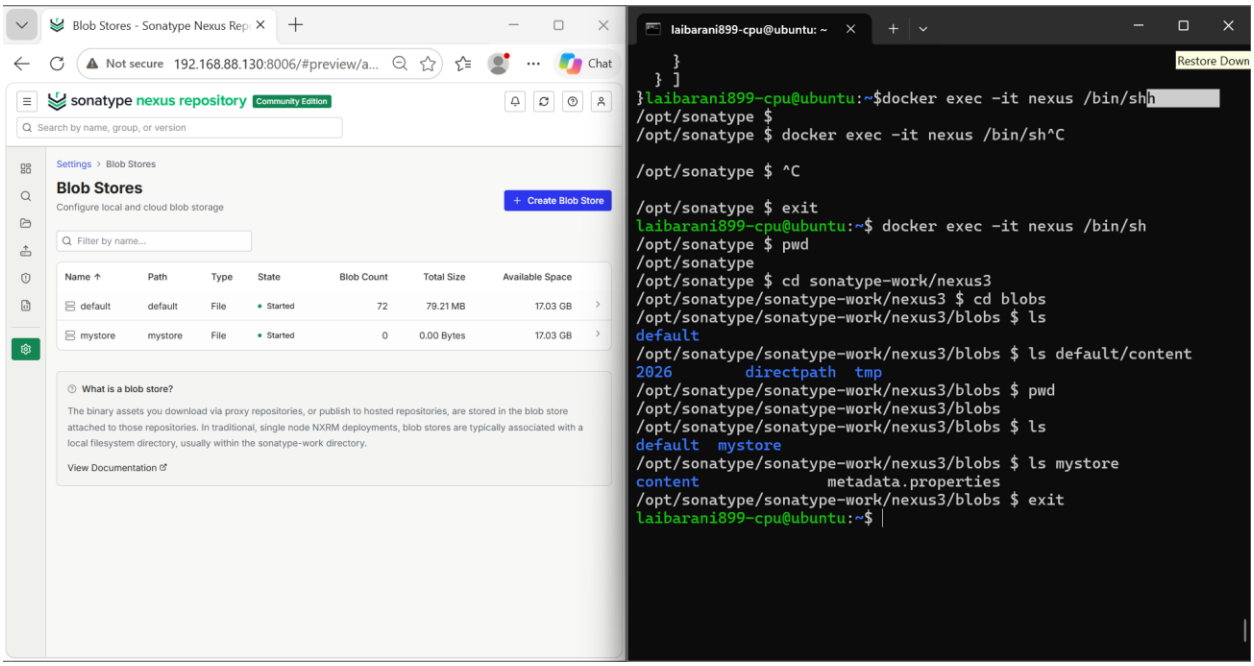
Verify the cleanup service schedule



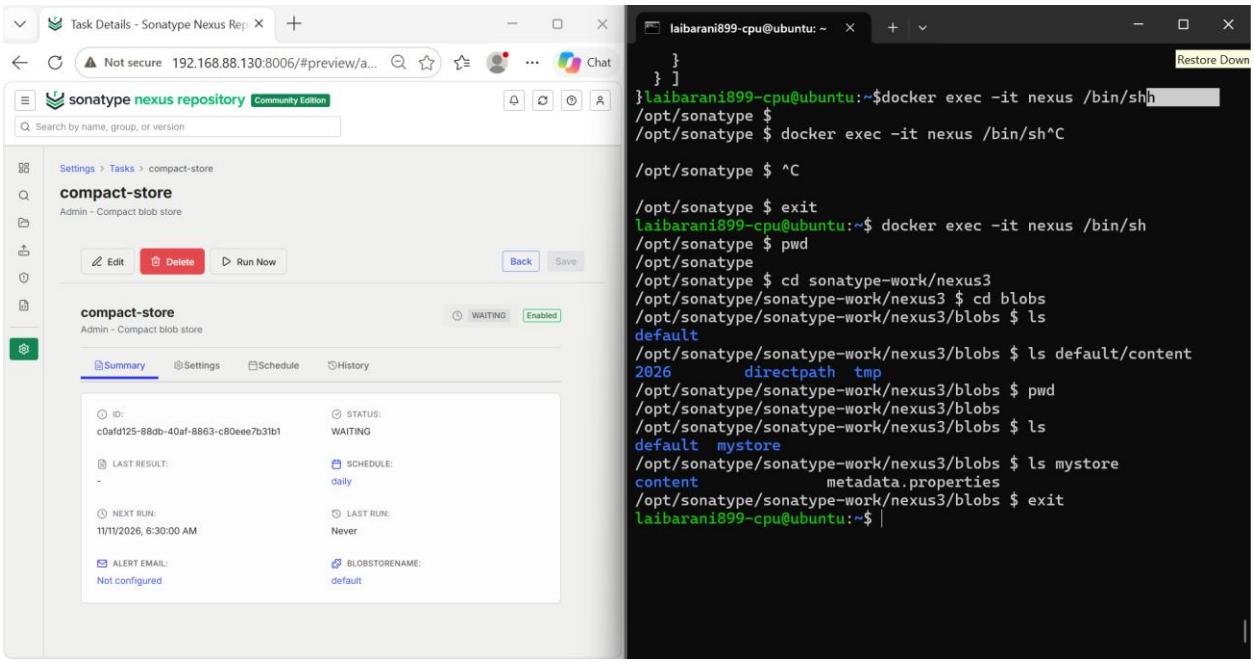
Run the cleanup service manually



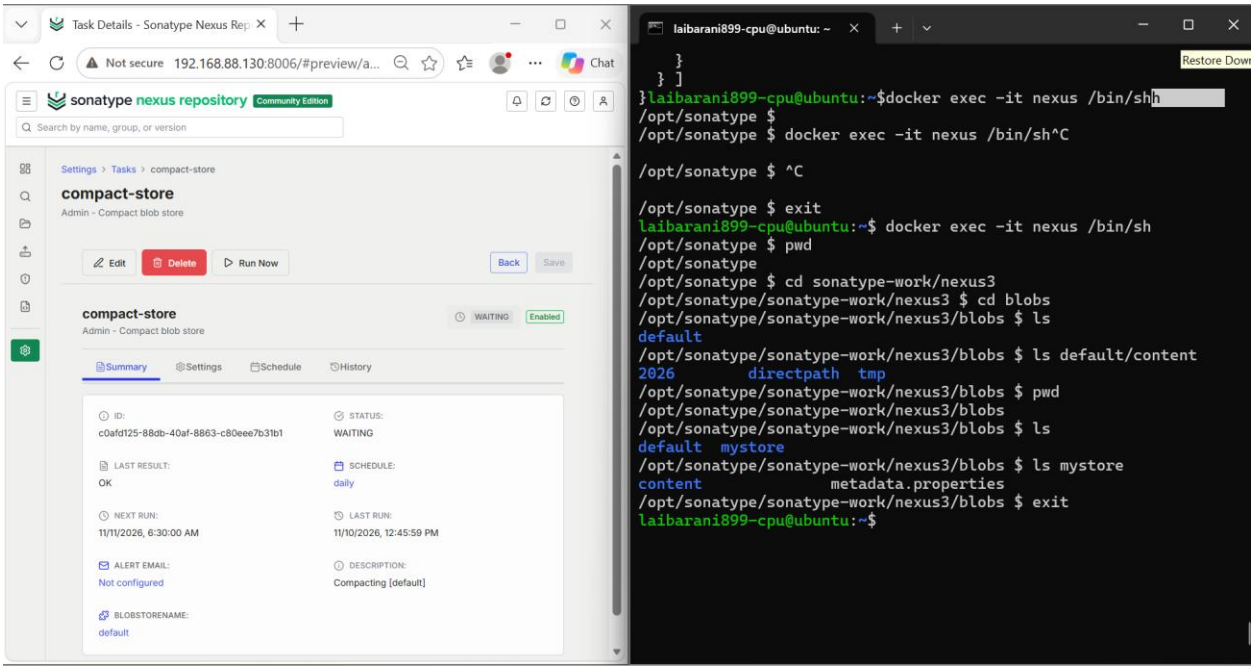
Verify that `maven-snapshots` is empty



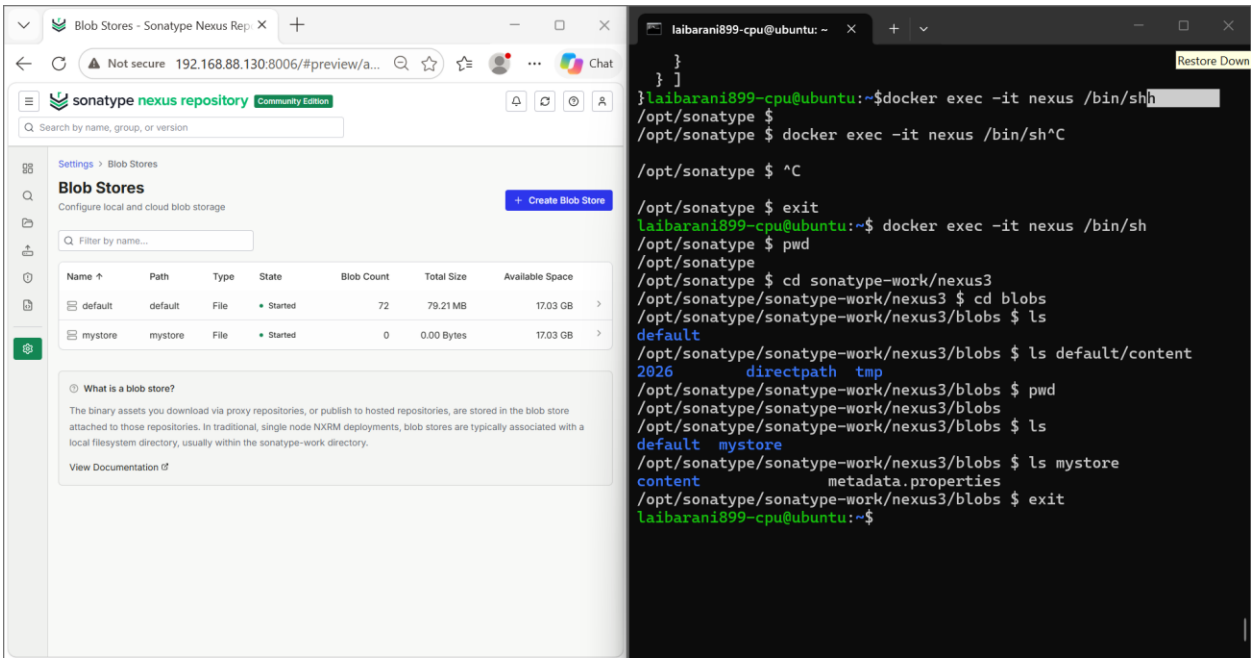
Check the blob store before compaction



Create the compact blob-store task



Run the compact task manually



Verify that blob-store space was reclaimed